

KEVIN FITZSIMMONS

Bioresource Engineering Graduate | McGill University

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Bioresource Engineering graduate relocating to Vancouver, BC with hands-on research experience in stormwater systems and urban ecology, and a decade of professional culinary experience cultivating precision, composure, and operational leadership. Core focus in pollution remediation, water systems engineering, and engineering for sustainability.

SELECTED EXPERIENCE

Capstone Design

BREE 490/495 – McGill University

📅 Jan – Dec 2025 📍 Montréal, QC

- Designed a zero-waste digestate management system for a Kentucky biogas facility processing whiskey distillery waste, integrating nutrient recovery across four process stages. Designed, built, and experimentally validated a custom photobioreactor to optimize microalgae-based nutrient removal from liquid digestate. Awarded the **CSABE | SCGAB Undergraduate Thesis Award (2025)**.

Research Assistant

IRBV, Université de Montréal

📅 May – Aug 2024 📍 Montréal, QC

- Authored the standardized field sampling protocol adopted across four countries (Canada, Sweden, Estonia, France) for BIORESTORM, an international initiative evaluating nature based solutions in urban stormwater systems.
- Conducted multi-parameter analysis of de-icing salt impacts on urban tree species, measuring water levels, soil conductivity, and mineral ion concentrations.

Sous Chef

Double's Late Night

📅 May – Aug 2022 📍 Montréal, QC

- Led kitchen operations for a new team; reduced food waste, developed seasonal menus with local ingredients, and implemented a composting program to meet municipal regulations.

Chef de Partie

Vin Mon Lapin

📅 Jun – Aug 2022 & Jun – Aug 2023 📍 Montréal, QC

- Supported operations at Canada's #1 ranked restaurant, executing market-driven menus while managing inventory and vendor coordination.

EDUCATION

B.Eng, Bioresource Engineering

McGill University

📅 2021 – 2025

B.Sc, Natural Resource Management

Univ. of British Columbia

📅 2010 – 2012

AWARDS



Undergraduate Thesis Award

CSABE | SCGAB (2025)



Stuart Horne Scholarship in Water Management

McGill University (2025)



VP, Fermenters & Brewers Club

McGill University (2023–24)



Eagle Scout Rank

Boy Scouts of America (2010)



William T. Hornaday Award

Boy Scouts of America (2010)

SKILLS

MATLAB

QGIS

AutoCAD

LaTeX

Fusion 360

3D Printing

Hydrological Modelling

SWAT

Water Quality & Field Sampling

Experimental Design

Technical Writing

Team Leadership

Class 5 Drivers License

RELEVANT COURSEWORK

Water Resources & Hydrology

- **BREE 217 — Hydrology and Water Resources:** Hydrologic cycle, precipitation and frequency analysis, soil water processes, evapotranspiration, surface runoff, unit hydrograph theory, and design of open channels and vegetated waterways.
 - **BREE 509 — Hydrologic Systems and Modelling:** Quantitative watershed modelling at point and catchment scale, including snowmelt, infiltration, soil water redistribution, subsurface drainage, and streamflow prediction using weather-driven physical models.
 - **BREE 416 — Engineering for Land Development:** Soil loss computation, conservation farming, reservoir and embankment design, water and sediment control structures, stream restoration, wetland design, irrigation, pumping systems, and drainage management.
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Water Quality & Pollution Remediation

- **BREE 533 — Water Quality Management:** Fate and transport of pollutants in rivers, lakes, and wetlands; soil nutrient cycling and leaching; agricultural drainage impacts; and numerical modelling for water quality improvement at real field sites.
 - **CHEE 591 — Environmental Bioremediation:** Role of microorganisms in natural attenuation and engineered bioremediation, and application of microbes in pollution control and environmental monitoring.
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Ecological Engineering & GIS

- **BREE 518 — Ecological Engineering:** Design and management of engineered ecosystems for service provision, with applications in constructed wetlands, green stormwater infrastructure, aquatic production systems, and environmental restoration. Included an international collaborative project with the University of Nebraska-Lincoln.
 - **BREE 529 — GIS for Natural Resource Management:** Spatial data capture, transformation, and display; spatial analysis methods applied to ecological datasets for natural resource management.
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Capstone Design: Biogas-ZE Digestate Management System

- **Process Design (BREE 490):** Developed a full-scale waste valorization system for a Kentucky biogas facility converting whiskey distillery waste into marketable outputs. These included recovered phosphorus fertilizer, treated water, and soil amendments, all while achieving net-zero emissions. Evaluated and selected technologies for each process stage based on energy efficiency, regulatory compliance, and commercial viability. Awarded the **CSABE | SCGAB Undergraduate Thesis Award (2025)**.
- **Experimental Design & Fabrication (BREE 495):** Designed and built a custom algae cultivation system from scratch to validate the nutrient removal component of the process design. Conducted controlled experiments demonstrating effective phosphorus removal from wastewater, and created an experimental design to optimize operating conditions for scale-up. Applied Fusion 360 CAD, 3D printing, and hands-on fabrication throughout.